Thinking, Fast and Slow, Formalized: When a Calibrated Decision Model Should Wake a Reasoning Model in Service and Software Systems

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Abstract

Kahneman's account of two modes of thought — a fast, automatic System 1 and a slow, effortful System 2 that is engaged only when needed — has become an engineering blueprint: "System One" decision models such as Jev return typed, calibrated judgments in half a second without generating text, while hybrid reasoning models such as Qwen3-8B can think at a controllable token budget. We give the interaction between such systems a mathematical form that tracks the psychological literature closely. Conflict detection and override become an exact escalation identity: the value of deliberation equals the rate at which the two systems disagree times System 2's win margin on those disagreements, the first factor observable without labels. Trustworthy intuition (Klein and Kahneman) becomes an act-band error guarantee under calibration, with a finite-sample rule for when calibration fails. Anchoring and rationalisation become an exact paired decomposition. Kahneman's theory of effort as a limited resource becomes an expected-value-of-control problem whose solution produces the familiar act/deliberate/give-up bands for any shape of thinking returns, including the inverse scaling recently observed in reasoning models. The *Noise* programme becomes a theorem that aggregating System-2 samples corrects noise but entrenches bias. We map these results to service triage, software engineering, requirements engineering, logistics exceptions, e-government and agentic software, and give a pre-registered evaluation protocol with a verified harness. No measurements of either model are reported; the paper is a formal and methodological contribution.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*.

Keywords

dual-process theory, System 1 and System 2, metacognition, LLM cascades, calibration, test-time compute, service systems, software engineering

1 Introduction

Kahneman's *Thinking, Fast and Slow* [18] popularised a distinction long studied in psychology [10, 33]: judgments arrive quickly and automatically from System 1, and a slower, effortful System 2 is recruited when something signals that the quick answer may be wrong. The distinction was never only descriptive. It carried a theory of *when* to think — effort is costly and limited [17], System 2 is a "lazy controller" that endorses intuition unless alarmed, and the alarm is itself an intuition about one's own confidence [35].

AI systems are now being built to this blueprint. Jev is presented by its developer as the first "System One" model: given a text *state* and typed questions (yes/no, choice, score), it returns probabilities and a calibrated confidence in about half a second and generates no text [16]. Qwen3-8B combines thinking and non-thinking modes in one model and lets the caller cap reasoning with a token budget [29]. Practitioners already pair them — "Jev routes and verifies things, while the LLM provides the language" [27] — in routers, judges and triage services. What is missing is a theory of the pairing that is precise enough to design and audit such services, and faithful enough to the psychology to justify the vocabulary.

This paper supplies one. Each result formalises a specific construct from Kahneman's programme and its successors (Table 2): conflict detection and override [9, 28], the conditions under which intuition can be trusted [20], anchoring and rationalisation [28, 37], effort allocation [17, 24, 32], and noise [21]. We then show how the resulting controller applies to problems central to this conference — service operations, software engineering, logistics and e-government — and give a pre-registered protocol for testing it. Because the psychology of dual processes has itself been shaken by failed replications [15], we report no results we could not verify: the authoring environment could reach neither model, so the empirical claims are stated as falsifiable hypotheses rather than findings.

Contributions. (1) A formal correspondence between Kahneman's constructs and quantities in a two-model decision system (Sec. 4). (2) Nine propositions, including an exact escalation identity, a shape-free theory of thinking budgets, and a noise-versus-bias theorem for self-consistency (Sec. 5). (3) Five interaction patterns tied to competing dual-process architectures (Sec. 6). (4) Applications and a worked example for service and software systems (Sec. 7). (5) A pre-registered protocol and a released harness (Sec. 8).

2 Kahneman's Two Systems

2.1 Two modes, one mind

Kahneman describes System 1 as operating "automatically and quickly, with little or no effort and no sense of voluntary control" and System 2 as allocating attention to effortful mental activities [18]. Earlier formulations contrasted associative and rule-based reasoning [33]; Evans and Stanovich [10] later argued that the defining property of Type 2 processing is reliance on working memory and hypothetical thinking, not speed as such. A central mechanism is *attribute substitution*: when a target question is hard, System 1 answers an easier related one [19]. Frederick's Cognitive Reflection

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Test (CRT) [11] operationalises the resulting failure — the bat-and-ball item elicits a fast, confident, wrong answer that only reflective override corrects.

2.2 Heuristics, confidence and their failures

Tversky and Kahneman's heuristics — representativeness, availability, and anchoring and adjustment — produce systematic biases [37]. Kahneman's "what you see is all there is" (WYSIATI) explains why confidence tracks the coherence of the story System 1 constructs rather than the quality of the evidence, producing an *illusion of validity* [18]. Whether intuitive confidence deserves trust was the subject of Kahneman and Klein's adversarial collaboration: intuitions are trustworthy only in environments with valid cues and adequate opportunity to learn them through feedback [20].

2.3 Metacognition: the alarm and the controller

If System 2 is engaged only when needed, something must decide when. Thompson et al. [35] showed that an initial answer arrives with a *feeling of rightness* whose strength predicts both rethinking time and the probability of changing the answer. De Neys [9] argued that reasoners often *detect* conflict between a heuristic answer and a normative one even when they go on to give the heuristic answer. Pennycook, Fugelsang and Koehler [28] integrated these findings in a three-stage model: initial responses are generated (Stage 1); conflict between them is or is not detected (Stage 2); and if analytic processing is engaged (Stage 3), it either *rationalises* the initial answer or *decouples* from it and overrides. Ackerman and Thompson [1] frame the whole as meta-reasoning — monitoring that feeds *control*, including the decision to stop thinking or give up.

2.4 Effort as a resource

Before the two-systems vocabulary, Kahneman's *Attention and Effort* [17] treated mental effort as a limited capacity allocated by an allocation policy. His later "law of least effort" states that when several routes reach the same goal, people gravitate to the least demanding [18]. Modern theory makes the allocation normative: the *expected value of control* weighs the payoff of engaging control against its intrinsic cost [32], and resource-rational analysis models cognition as optimal use of bounded computation [24]. One strand of this literature, ego depletion, failed a large multi-laboratory registered replication [15]; we therefore rely only on the cost–benefit formulation, which does not depend on it.

2.5 Noise and decision hygiene

Noise [21] separates *bias* (systematic error) from *noise* (unwanted variability), including occasion noise — the same judge answering differently on different occasions — and recommends *decision hygiene*: collect independent judgments before they can influence each other, then aggregate. Averaging independent estimates from one person improves accuracy, the "crowd within" [38].

3 Machine Instantiation

Table 1 contrasts the two systems. Jev's design reads like a specification of an idealised System 1: parallel typed judgments, no generation, and a confidence trained to be *calibrated* — a property human System 1 lacks. Qwen3-8B's thinking mode is serial, effortful in tokens and latency, and adjustable in depth, with performance reported to scale with the reasoning budget [29]. Two recent findings on reasoning models echo the psychology: models *overthink* easy problems [6], and on some tasks accuracy *falls* as reasoning lengthens — inverse scaling in test-time compute [12]. Related AI work includes the SOFAI architecture of fast and slow solvers under a metacognitive module [4], cost-aware LLM cascades [5], selective classification with risk guarantees [13], threshold calibration as multiple testing [2], difficulty-adaptive test-time compute [34], and self-consistency sampling [40]. In a companion study on tree ensembles [23] the fast/slow mapping failed as a speed claim; here the cost gap is real.

Table 1: The two systems (vendor documentation [16, 29, 30]).

	System 1: Jev	System 2: Qwen3-8B
Output	typed answer, probability for every option, confidence	free text; answer parsed from final line
Generation	none	up to 32k tokens (thinking)
Latency, cost	∼0.5 s; $0.042 per million input tokens	GPU-bound; grows with budget B
Confidence	trained for calibration	verbalised confidence often overconfident [41]
Weak spots	numbers, dates, adversarial text	overthinking, inverse scaling [6, 12]

4 A Formal Correspondence

Let a query consist of a text state, a typed question with options $\mathscr{A}$ ($|\mathscr{A}| = K$) and gold answer y. System 1 returns a_1, a probability vector p and confidence $c = \max_k p_k$; System 2 with thinking budget B returns a_2 (B=0 is non-thinking mode). A controller chooses to *act* (release a_1), to *deliberate* with budget B (release a_2), or to fall back. Z denotes the set of deliberated queries. Table 2 lists the correspondences developed below.

5 Mathematics of the Interaction

5.1 Conflict and override: the escalation identity

PROPOSITION 5.1 (ESCALATION IDENTITY). *Under 0–1 loss, the accuracy gained on Z by releasing a_2 instead of a_1 is*

$$G(Z) = M_Z\,(W_2 - W_1), \qquad M_Z = P(x{\in}Z,\ a_1{\neq}a_2),$$

where $W_k = P(a_k{=}y \mid x{\in}Z, a_1{\neq}a_2)$. For K=2, $G(Z) = M_Z(2W_2 - 1)$.

PROOF. $G(Z) = \mathbb{E}[\mathbf{1}_Z(\mathbf{1}\{a_2{=}y\} - \mathbf{1}\{a_1{=}y\})]$; the integrand vanishes when $a_1{=}a_2$. Condition on disagreement. For K=2 exactly one of two disagreeing answers is correct. □

COROLLARY 5.2 (WHERE VERSUS WHETHER). *M_Z is computable on unlabelled traffic; $W_2 - W_1$ requires labels.*

Interpretation. The identity separates De Neys's conflict detection from Pennycook's override. M_Z is the machine analogue of Stage 2: it registers that the two systems disagree, and it can be measured without knowing who is right. $W_2 - W_1$ is the quality of Stage 3

Table 2: Kahneman's constructs and their formal counterparts in a two-model decision system.

Psychological construct	Formal object	Result and prediction
Feeling of rightness [35]; cognitive ease	calibrated confidence c	Prop. 5.3: error of acting on $c \geq \tau$ is at most $1-\tau$
Trustworthy intuition requires valid cues and feedback [20]	calibration on exchangeable traffic; labelled feedback set	Remark 1: a few hundred labels suffice to certify the act band; fails out of distribution
WYSIATI, illusion of validity [18]	confident errors, $P(a_1 \neq y \mid c \geq \tau) > 1-\tau$	act-band failure on adversarial or lure items (H1′, H8)
Conflict detection [9]; Stage 2 [28]	disagreement mass $M_Z = P(Z, a_1 \neq a_2)$	Prop. 5.1: observable without labels
Override / decoupling [28]	win margin $W_2 - W_1$ on disagreements	Prop. 5.1: deliberation helps iff System 2 wins most disagreements
Anchoring [37]; rationalisation [28]	paired transitions blind $\to$ informed	Prop. 5.4: anchor, reactance and drift terms, exact
Decision hygiene [21]; consider the opposite [25]	blind escalation; blind-then-reconcile	Cor. 5.5: pure anchoring scales the benefit by $1-\alpha$
Effort as limited capacity [17]; EVC [32]; resource rationality [24]	$\max_B s(c)h(B) - \lambda B$	Props. 5.6–5.7: act/deliberate/give-up bands
Law of least effort [18]	zero budget above a confidence threshold	act band edge depends only on $\rho^\star = \sup h(B)/B$
Overthinking, rumination	flip rate r; inverse scaling	budget capped at the peak of h (Prop. 5.7a)
Knowing when to give up [1]	fallback when post-deliberation error $> \varepsilon$	Props. 5.9–5.10: Bayes action, conformal sets
Occasion noise; crowd within [21, 38]	majority of k System-2 samples	Prop. 5.11: voting corrects noise, entrenches bias

decoupling. A deliberation that ends in agreement — System 2 endorsing System 1, Kahneman's lazy controller at work — contributes exactly zero, however much it cost.

5.2 Trustworthy intuition: the act band

PROPOSITION 5.3 (ACT-BAND GUARANTEE). *If System 1 is calibrated, $P(a_1{=}y \mid c) = c$, then for any threshold τ, $P(a_1{\neq}y \mid c \geq \tau) = \mathbb{E}[1-c \mid c \geq \tau] \leq 1-\tau$.*

PROOF. The band is a function of c, so calibration holds on it; apply the tower property. □

Remark 1. Calibration is an aggregate property [16], and Kahneman and Klein's conditions for valid intuition [20] have a direct statistical reading: the act band is trustworthy only on traffic resembling the feedback on which it was certified. Given n labelled items, scan τ from high to low and stop at the first threshold whose Clopper–Pearson upper bound on act-band error exceeds ε. Fixed-sequence testing then bounds act-band error by ε with probability $\geq 1-\delta$ under exchangeability, whether or not Jev is calibrated [2]. WYSIATI predicts where this breaks: text engineered to be coherent but misleading (prompt injection, distractors, CRT-style lures) can make a System-1 model confidently wrong, and no threshold on its own confidence will catch that.

5.3 Anchoring and rationalisation: paired transitions

Showing System 2 the pair (a_1, c) can reshape its entire chain of thought, not merely swap its final answer. We make no assumption about that mechanism and compare outputs in a *paired* design: on each escalated query System 2 answers blind (b) and informed (i).

PROPOSITION 5.4 (PAIRED TRANSITION DECOMPOSITION). *Partition escalated queries into* stable *($i{=}b$),* anchor *($b{\neq}a_1$, $i{=}a_1$),* reactance *($b{=}a_1$, $i{\neq}a_1$) and* drift *($b{\neq}a_1$, $i \notin \{a_1, b\}$, only for $K{>}2$). Then*

$$G^{\text{inf}} - G^{\text{blind}} = \textstyle\sum_T P(T)\big(P(i{=}y \mid T) - P(b{=}y \mid T)\big),$$

with a zero stable term, for any effect of the hint on the reasoning trace.

PROOF. $G^{\text{inf}} - G^{\text{blind}} = \mathbb{E}_Z[\mathbf{1}\{i{=}y\} - \mathbf{1}\{b{=}y\}]$ and the four types partition Z. □

COROLLARY 5.5 (PURE ANCHORING). *If only anchor transitions occur, at a rate α independent of correctness given blind disagreement, then $G^{\text{inf}} = (1-\alpha)G^{\text{blind}}$.*

Interpretation. Anchor transitions are Stage-3 *rationalisation* in Pennycook et al.'s sense: the analytic process arrives where the initial response already was. Reactance, where a hint pushes System 2 away from a correct a_1, and drift, where it derails reasoning to a third option, are separate terms rather than unmodelled noise. Corollary 5.5 is the formal case for Kahneman et al.'s decision hygiene: obtain the independent judgment first. The blind-then-reconcile pattern (Sec. 6) does exactly that, confining any anchoring to a short, auditable "consider the opposite" step [25].

5.4 Effort as a resource: how much to think

Kahneman's allocation policy [17] and the expected value of control [32] weigh the benefit of control against its cost. In our setting the benefit of thinking budget B on a query with confidence c is $s(c)h(B)$ and the cost is λB.

PROPOSITION 5.6 (SATURATING RETURNS). *Let $K{=}2$ and $d = 2(1-c)$. Suppose deliberation fixes a System-1 error with probability $(1-d^\gamma)h(B)$ and overturns a correct answer with probability $r\,h(B)$, with $h(B) = 1-e^{-B/\beta}$. The net value $s(c)h(B) - \lambda B$, where $s(c) = [(1-c) - rc](1-d^\gamma)$, is maximised by $B^* = \beta \ln(s/\lambda\beta)$ if $s > \lambda\beta$ and $B^* = 0$ otherwise. No thinking is bought for $c \geq (1-\lambda\beta)/(1+r)$; $s(c) \to 0$ as $c \to \frac{1}{2}$; and B^* peaks at intermediate confidence.*

PROOF. The objective is concave in B; solve $s\,e^{-B/\beta}/\beta = \lambda$. □

The three regions are Kahneman's: the law of least effort (act when confident), effortful engagement at intermediate confidence,

Table 3: Optimal budgets under five return curves (numerical; model properties).

$h(B)$	act $c\geq$	give up $c\leq$	peak B^*	min $B^*>0$	kept
saturating	0.90	0.57	2.76k	57	100%
log-linear	0.90	0.59	2.62k	10	97%
early-flat	0.91	0.58	0.55k	8	71%
threshold	0.85	0.57	5.52k	3.74k	45%
inverted-U	0.90	0.56	2.07k	10	97%

"kept": value of the saturating closed-form budget relative to the true optimum.

and a metacognitive decision to give up [1] on queries that are intrinsically ambiguous, where System 1's error is highest but deliberation cannot help. The overthinking ratio r sets the upper edge.

Real thinking returns need not be smooth: they can flatten almost at once, require a threshold of reasoning before any benefit, or decline as reasoning lengthens [6, 12]. We keep only separability.

PROPOSITION 5.7 (SHAPE-FREE STRUCTURE). *Let h be any function with $h(0)=0$, $\bar{B}$ the smallest maximiser of h, and $\rho^\star = \sup_{B>0} h(B)/B$. Then: (a) every optimal budget lies in $[0,\bar{B}]$; (b) the largest optimal budget is nondecreasing in $s(c)$; (c) $B^*(c)>0$ iff $s(c)>\lambda/\rho^\star$, so with s unimodal the deliberate region is an interval and the curve affects the band edges only through $\rho^\star$; (d) if $h(B)/B$ peaks at $B_{\tan}>0$, no positive budget below $B_{\tan}$ is optimal.*

PROOF. (a) Beyond $\bar{B}$ returns do not rise and cost does. (b) On $[0,\bar{B}]$ the objective has increasing differences in (B,s); apply monotone comparative statics [26, 36]. (c) $B>0$ beats $B=0$ iff $s\,h(B)>\lambda B$ for some B. (d) At an optimum $B<B_{\tan}$, $h(B)/B\geq\lambda/s$; moving to $B_{\tan}$ adds value $s[h(B_{\tan})-h(B)]\geq(B_{\tan}-B)\,s\,h(B)/B\geq\lambda(B_{\tan}-B)$, at least its extra cost. □

PROPOSITION 5.8 (MISSPECIFICATION REGRET). *If $\hat{g}$ estimates the net accuracy gain g with $\sup_B|g-\hat{g}|\leq\eta$, the plug-in budget $\arg\max_B \hat{g}(B)-\lambda B$ loses at most 2η.*

PROOF. $V(B^*)-V(\hat{B})\leq[V(B^*)-\hat{V}(B^*)]+[\hat{V}(\hat{B})-V(\hat{B})]\leq 2\eta$. □

Behaviour under realistic scaling. Table 3 and Fig. 1 evaluate the exact optimum for five return curves spanning reported behaviour, using the parameters of Prop. 5.6 (r=0.1, γ=2, $\lambda\beta$=0.01, β=1024, give-up tolerance ε=0.35). These are properties of the model, not measurements. The bands barely move: acting begins at $c\in[0.85,0.91]$ and giving up ends at $c\in[0.56,0.59]$ for every curve. Magnitudes move tenfold. Under threshold returns a minimum effective budget of 3.7k tokens appears, so the policy becomes all-or-nothing — a cascade. A closed form fitted for saturating returns keeps 97–100% of the achievable value under three curves but only 71% (early-flat) and 45% (threshold). Decisions about *whether* to think are robust; decisions about *how much* are not. When curves differ by task, a single pooled policy keeps 70% of the value of a task-conditioned one; Jev can supply the task label through an auxiliary typed question answered in the same call.

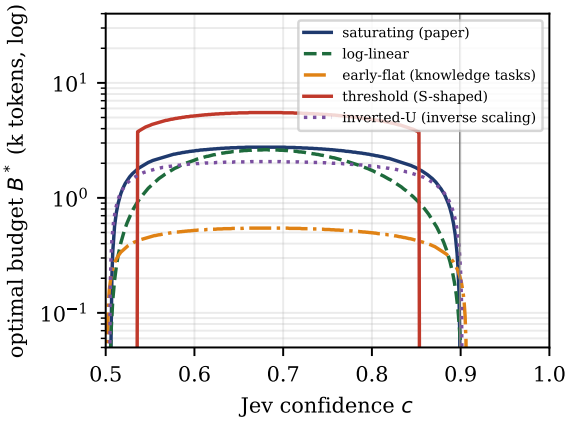


Figure 1: Optimal thinking budget $B^*(c)$ under five return curves (log scale). Band edges are nearly invariant; magnitudes and the minimum effective budget are not.

5.5 Giving up without a human

Many real-time services cannot route hopeless queries to a person. Jev returns a probability for every option, which supports two principled fallbacks.

PROPOSITION 5.9 (CALIBRATED BAYES ACTION). *If $P(y{=}k\mid p)=p_k$ for all k, then for any loss $L(a,y)$ over an action set that may include a conservative no-op ("hold", "safe queue"), the rule $a^*(p)=\arg\min_a\sum_k p_k L(a,k)$ minimises expected loss among all rules that depend on the query only through p.*

PROOF. $\mathbb{E}[L(\delta(p),y)]=\mathbb{E}[\sum_k p_k L(\delta(p),k)]$ by calibration; minimise pointwise. □

PROPOSITION 5.10 (CONFORMAL ANSWER SETS). *With n exchangeable labelled items and $\hat{q}$ the $\lceil(n{+}1)(1{-}\alpha)\rceil$-th smallest score $1-p_y$, the set $C(x)=\{k:1-p_k(x)\leq\hat{q}\}$ satisfies $P(y\in C(x))\geq 1-\alpha$ [31].*

Prop. 5.9 needs calibration of the *full* vector, which is stronger than confidence calibration and is tested in the protocol. It also marks a deliberate departure from human System 2, whose choices under risk show loss aversion [22]: the machine fallback should encode the organisation's actual losses, not reproduce a descriptive bias. The fallback order is: a human if one is available; otherwise information acquisition (retrieval, a clarifying question) when affordable; otherwise the Bayes action, or a minimax action over the conformal set.

5.6 Noise and the crowd within

Sampling System 2 several times and taking the majority (self-consistency [40]) is the machine version of the crowd within [38]. Kahneman et al.'s distinction between bias and noise determines when it helps.

PROPOSITION 5.11 (AGGREGATION CORRECTS NOISE, ENTRENCHES BIAS). *Let $K{=}2$ and let System 2 answer query i correctly with probability p_i on each of k (odd) independent runs. The*

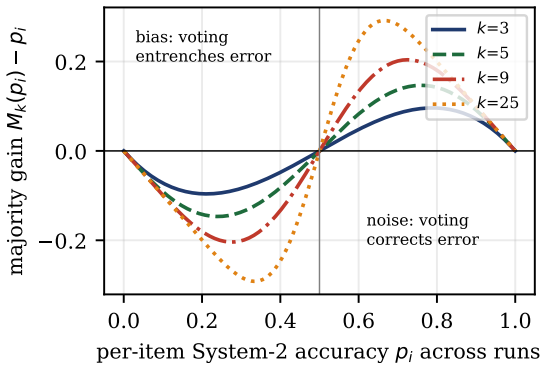


Figure 2: Prop. 5.11: gain from majority voting over k System-2 runs as a function of per-query accuracy. Voting corrects noise ($p_i > \frac{1}{2}$) and entrenches bias ($p_i < \frac{1}{2}$).

majority is correct with probability $M_k(p_i) = P(\mathrm{Bin}(k, p_i) > k/2)$, and $M_k(p) - p$ has the sign of $p - \frac{1}{2}$, with $M_k(p) \to \mathbf{1}\{p > \frac{1}{2}\}$ as $k \to \infty$. Hence the expected gain from voting, $\mathbb{E}_i[M_k(p_i) - p_i]$, is positive on queries where System 2 is noisy but right on average and negative where it is systematically wrong.

PROOF. $M_k(p) + M_k(1-p) = 1$, so $M_k(\frac{1}{2}) = \frac{1}{2}$; M_k is increasing, with $M_k'(p) = k\binom{k-1}{(k-1)/2}[p(1-p)]^{(k-1)/2}$ exceeding 1 near $\frac{1}{2}$ and vanishing at the ends. Hence $M_k(p) - p$ has a single interior root at $\frac{1}{2}$ on $(0,1)$ and takes the sign of $p - \frac{1}{2}$; the limit is the law of large numbers. This is the per-query form of Condorcet's jury theorem [14]. □

Fig. 2 plots the effect. Voting is decision hygiene against occasion noise, but on items where System 2 has a systematic misconception — the machine analogue of a biased judge — more samples make the error more certain. Samples from one model are also correlated, which shrinks the effective k. The protocol therefore runs a *noise audit*: the rate at which repeated runs on the same query disagree.

6 Interaction Patterns

Dual-process theories disagree about architecture. *Default-interventionist* accounts have System 1 answer first and System 2 intervene when needed [10]; *parallel-competitive* accounts run both and resolve conflict [33]. Both have machine counterparts (Fig. 3). **P1 Cascade** (default-interventionist): deliberate at a fixed budget when $c < \tau$, with τ certified as in Remark 1. **P2 Budgeted deliberation**: deliberate with a task-conditioned budget estimated nonparametrically on a budget grid (Prop. 5.8 bounds its regret); the closed form of Prop. 5.6 is a baseline. **P3 Disagreement probe** (parallel-competitive): run Qwen3-8B in non-thinking mode, which costs tens of tokens, alongside Jev, and think only on disagreement or when $c < \tau$. This estimates the label-free conflict signal M directly and can catch confident System-1 errors that a confidence threshold misses — De Neys's conflict detection without low confidence. **P4 Jev as verifier**: when Qwen3-8B overturns Jev, Jev judges whether the new answer is supported by the passage and reasoning; if not, the override is reverted. This inverts the usual direction and targets Stage-3 rationalisation by System 2 itself. **P5 Blind-then-reconcile**: System 2 reasons blind, then sees (a_1, c) in a short second turn and may revise only on finding a concrete error — decision hygiene as an interaction protocol.

Algorithm 1 combines the patterns into one controller.

Algorithm 1: Dual-process controller (one query)

Input: query x; certified threshold τ; budget table $\hat{B}(c, \text{task})$; give-up tolerance ε; loss L; flags *probe*, *reconcile*, *verify*

$(a_1, p, c, \text{task}) \leftarrow \text{Jev}(x)$; `// main and task questions in one call`
if $c \geq \tau$ **then return** a_1 ; `// act: law of least effort`
;
$B \leftarrow \hat{B}(c, \text{task})$;
if *probe* **then**
  $a_0 \leftarrow \text{Qwen}(x, 0)$; **if** $a_0 = a_1$ ***and*** *B small* **then return** a_1;
if $B = 0$ ***or*** *post-error*$(c, B) > \varepsilon$ **then return** Fallback(p) ; `// give up`
;
$b \leftarrow \text{Qwen}(x, B)$ blind;
if *reconcile* **then** $b \leftarrow \text{Reconcile}(b, a_1, c)$;
if *verify* ***and*** $b \neq a_1$ **then**
  if *Jev judges b unsupported with high confidence* **then return** a_1;
return b;

Fallback(p): human if available; else acquire information if affordable and restart; else $\arg\min_{a \in \mathscr{A} \cup \{\text{hold}\}} \sum_k p_k L(a, k)$ or a minimax action over the conformal set.

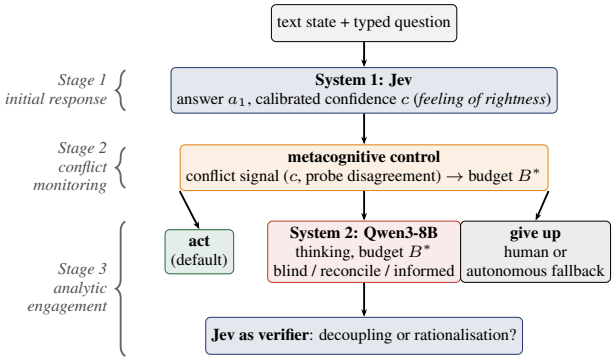


Figure 3: Architecture mapped onto Pennycook et al.'s three-stage model [28]. Stage 1: Jev's answer and feeling of rightness. Stage 2: conflict monitoring sets the budget. Stage 3: System 2 decouples or rationalises, audited by Jev.

7 Applications in Service and Software Systems

The controller fits any service in which many decisions are cheap to make, a minority deserve deliberation, and mistakes have asymmetric costs. Table 4 summarises six domains within the scope of this conference.

Table 4: Applications. Each row names System 1's typed questions, System 2's deliberative task, the dominant asymmetric loss, the fallback, and the dual-process risk to monitor.

Domain	System 1 (Jev) typed questions	System 2 (Qwen3-8B) task	Costly error	Fallback	Risk to monitor
Customer service / CRM	category; urgency; churn risk; refund eligibility	resolution plan for complex, multi-issue tickets	urgent ticket mis-routed	priority queue (hold)	anchoring on System-1 category
Software engineering	PR risk tier; test failure flaky vs. real; page on-call?	reason over diff or logs; propose root cause	missed security regression	block merge; page human	rationalised overrides (use P4)
Requirements engineering	functional vs. non-functional; ambiguous? testable?	rewrite ambiguous requirement; derive acceptance tests	ambiguity shipped to build	conformal set; ask stakeholder	overthinking clear items
Logistics / port operations	delay cause; customs hold; dangerous-goods flag	re-plan berth or route under constraints	unsafe cargo released	hold for inspection	WYSIATI on coherent but false notices
E-government services	eligibility pre-screen; document complete?	check against regulation text	wrongful denial	mandatory human review	bias vs. noise across groups
Agentic software	tool result relevant? injected instruction?	multi-step planning	context poisoning	drop result; stop agent	confident errors on adversarial text

Service operations. In a help desk, Jev answers routing questions for every ticket and Qwen3-8B drafts plans only for tickets in the deliberate band. The loss matrix is explicit: a mis-routed urgent ticket costs far more than a short delay, so the Bayes fallback (Prop. 5.9) will often prefer the "priority queue" no-op to a confident but uncertain category. *Software engineering.* Pull-request risk routing, CI-failure triage and incident paging are high-volume typed decisions; community tools already route pull requests and model tiers with Jev [16]. Deliberation over diffs and logs is expensive and prone to fluent but wrong root-cause stories, which the verifier (P4) targets. *Requirements engineering.* Classifying requirements and flagging ambiguity is a typed task; conformal sets expose requirements whose classification is genuinely uncertain, which should be clarified with stakeholders rather than guessed. *Logistics.* Exception notices for shipments and vessel calls (delay causes, customs holds, dangerous-goods declarations) arrive at volume; the natural fallback is to hold for inspection. The WYSIATI risk is concrete: a well-formed but false notice can make System 1 confidently wrong, which is why the probe (P3) runs in parallel. *E-government.* For decisions affecting rights and entitlements, human review must stay first in the fallback order regardless of cost, and disparities should be audited separately as bias (systematic, by group) and noise (inconsistency) in Kahneman et al.'s sense. *Agentic software.* Judging whether each tool result is relevant or contains an injected instruction before it enters an agent's context is a System-1 task at high frequency; deliberation is reserved for planning.

Worked example (illustrative parameters, not measurements). Take a binary triage service whose System-1 confidence is distributed as $c = 0.5 + 0.5\,U$, $U \sim \mathrm{Beta}(3,1)$, and use the saturating model of Prop. 5.6. Acting on System 1 alone yields accuracy 0.875. Thinking at 4,096 tokens on every query yields 0.895: the gain on uncertain queries is partly cancelled by overthinking on confident ones. Table 5 shows the budgeted controller across three price regimes. At $\lambda\beta$=0.01 it decides about half the queries fast, deliberates on the other half, gives up on 0.2%, and reaches 0.915 on decided queries at about 1,000 tokens per query — higher accuracy than thinking on everything at a quarter of the tokens. Raising the price of thought shifts queries from deliberation to acting, as the law of least effort predicts.

Table 5: Worked example: shares, mean thinking tokens per query, and accuracy on decided queries under three prices of thought (illustrative).

Policy	act	deliberate	give up	tokens	accuracy
System 1 only	100%	—	—	0	0.875
Think on all, B=4096	—	100%	—	4,096	0.895
Budgeted, $\lambda\beta$=0.003	47%	53%	0.2%	1,658	0.918
Budgeted, $\lambda\beta$=0.01	49%	51%	0.2%	1,015	0.915
Budgeted, $\lambda\beta$=0.03	56%	43%	0.4%	478	0.906

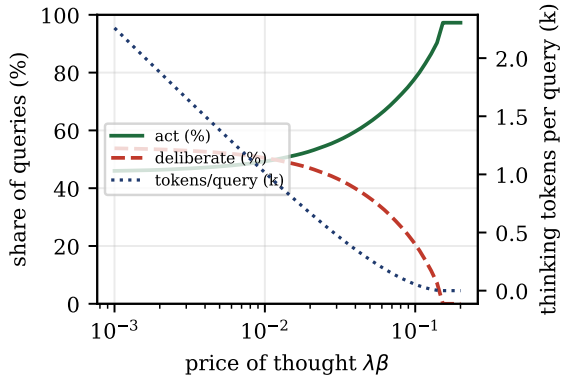


Figure 4: Price of thought (worked example): as thinking becomes more expensive, queries move from deliberation to acting and tokens per query fall.

Fig. 4 traces the whole frontier: tokens per query fall from about 2,300 to zero as the price of thought rises, while accuracy on decided queries declines gracefully from 0.92 towards the System-1 level.

7.1 Deployment and quality assurance

Service levels. Jev runs on every query; Qwen3-8B only on the deliberate share, so tail latency is a mixture whose weight is read off Fig. 4, and capacity planning reduces to choosing the price of thought. *Failure handling.* When Jev errors or returns an out-of-schema answer, high-stakes services should fail *closed*, to the fallback rather than to a default route — the convention already adopted by community Jev routers. *Keeping intuition trustworthy.* Kahneman and Klein's feedback condition becomes an operational duty. The act band is certified on data from the past; to keep it valid, audit a random sample of act-band decisions and run Wald's sequential probability ratio test [39] of "error $\leq \varepsilon$" against "error $\geq \varepsilon + \Delta$", recertifying τ on rejection. Corollary 5.2 adds a label-free early warning: the disagreement rate M between Jev and the non-thinking probe can be tracked on all traffic, and a shift in M signals distribution drift before any labels arrive. *Versioning.* Thresholds, budget tables and conformal quantiles belong to a specific pair of model versions and must be recertified on any upgrade.

8 Pre-Registered Evaluation Protocol

Tasks. T1: BoolQ [7] validation (3,270 yes/no items), a Jev strength. T2: GSM8K [8] test (1,319) recast as four-option choice with distractors fixed in advance, targeting Jev's documented weakness with arithmetic. T3: CRT-style lure items [11]: newly written isomorphs with fresh numbers and wording, to avoid contamination, in which the intuitive answer is wrong, plus distractor-laden items of the kind that produce inverse scaling [12]. *Design.* 30% of each task is used for calibration (threshold certification; nonparametric budget curves on the grid $B \in \{0, 128, \ldots, 16\text{k}\}$, per task and confidence bin), 70% for test. All five patterns run; escalated items are run blind, informed and blind-then-reconcile in a paired design; a subset is run five times for the noise audit. Qwen3-8B uses the vendor-recommended sampling settings [30]. *Measures.* Accuracy on decided queries; give-up rate; M, W_1, W_2 by confidence band; act-band error with Clopper–Pearson bounds; transition shares; classwise calibration; conformal coverage and set size; occasion noise and majority gain; tokens, latency and cost.

Harness. The released code isolates API calls behind a Jev client and an OpenAI-compatible Qwen3-8B client, and implements every pattern, the nonparametric budget policy, the paired decomposition, the noise audit and both fallbacks. Tested against mock systems with known properties, the identities of Props. 5.1 and 5.4 hold to machine precision, the nonparametric policy recovers the bands, conformal coverage matches its target, and majority voting helps only under independent runs. These checks establish only that the code computes what the paper defines. Jev's request schema must be verified against current documentation before use.

9 Discussion

Where the analogy holds. The correspondence is structural rather than metaphorical: each construct in Table 2 is defined operationally and measured by a quantity the propositions constrain. The strongest parallels are conflict detection as label-free disagreement, effort allocation as expected value of control, and decision hygiene as blind escalation. *Where it breaks.* Human System 1 is not calibrated; Jev is trained to be, which is precisely why Prop. 5.3 is useful and also why its failure modes (WYSIATI on adversarial text) deserve dedicated tests. Human System 2 is not a separate agent with its own prior, whereas Qwen3-8B is, so "decoupling" here means disagreement between two models, not revision within one mind. A model's thinking tokens are not deliberation in the psychological sense, and we claim no equivalence of mechanism, only of decision structure. Finally, the loss-aversion contrast shows that the formalisation is normative: it tells a service how it *should* allocate deliberation, not how people do.

What the formalism predicts about people. The results are normative for machines but yield testable predictions for the psychology they borrow from. (i) The escalation identity implies that studies of reflection should report the rate of detected conflict and the success of override *separately*, since the benefit of reflection is their product; CRT scores conflate them. (ii) The budget results predict that rethinking time is *non-monotone* in the feeling of rightness: it should fall as confidence rises, as Thompson et al. found [35], but also fall again for items judged hopeless, where people give up [1]. (iii) Prop. 5.11 predicts that averaging a person's repeated answers helps on items they get right more often than not, and harms on items where a systematic lure dominates, such as CRT items; the crowd within should cure noise, not bias. (iv) Prop. 5.4 suggests that advice-taking studies [3] use paired designs that separate anchoring from reactance. *Limitations.* No empirical results. Props. 5.6–5.7 assume separable returns in confidence and budget; Props. 5.3 and 5.10 require exchangeable traffic; Prop. 5.9 requires classwise calibration; Prop. 5.11 assumes independent runs, which samples from one model are not.

Table 6: Pre-registered hypotheses and falsification criteria.

	Hypothesis	Falsified if
H1	T1: act-band error $\leq \varepsilon$ at the certified τ	error CI above ε
H1′	T2: Jev overconfident; act-band error $> 1-\tau$	error within $1-\tau$
H2	$W_2 > W_1$ in the middle band only; $W_2 \leq W_1$ in the top band	top-band $W_2 - W_1$ CI above 0
H3	Anchor share $>$ reactance share; informed worse than blind; reconcile has lower anchor share than informed	any inequality reversed with CI excluding 0
H4	Nonparametric B^* is 0 in the top band, positive in a middle interval; curves differ by task	$B^* > 0$ in top band, or no task difference
H5	Probe disagreement predicts thinking-mode disagreement (AUROC > 0.7)	AUROC ≤ 0.7
H6	Verifier cuts harmful overrides $\geq$ 30%, keeps $\geq$ 80% of beneficial ones	either missed
H7	Conformal coverage $\geq 1-\alpha-0.02$; Bayes action beats always-act under the stated loss	either missed
H8	T3: Jev confidently wrong on lures; probe catches more lure errors than any confidence threshold	threshold catches as many
H9	Voting gains confined to items with $p_i > \frac{1}{2}$; losses where $p_i < \frac{1}{2}$	gains where $p_i < \frac{1}{2}$

10 Conclusion

Kahneman's two systems can be given an exact mathematical form when System 1 is a calibrated decision model and System 2 a budgeted reasoning model. The value of deliberation is the rate of conflict times the quality of override; conflict is visible without labels, override quality is not. Calibrated confidence certifies when intuition can be trusted. Paired comparison measures anchoring and rationalisation without modelling them. Effort allocation produces act, deliberate and give-up bands for any shape of returns, but how much to think must be learned per task. Aggregating samples corrects noise and entrenches bias. These results give service and software engineers a principled way to decide when a fast system should wake a slow one — and a protocol that can prove them wrong.

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References

[1] R. Ackerman and V. A. Thompson. 2017. Meta-reasoning: Monitoring and control of thinking and reasoning. *Trends in Cognitive Sciences* 21, 8 (2017), 607–617.
[2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei. 2021. Learn then test: Calibrating predictive algorithms to achieve risk control. arXiv:2110.01052.
[3] S. Bonaccio and R. S. Dalal. 2006. Advice taking and decision-making: An integrative literature review, and implications for the organizational sciences. *Organizational Behavior and Human Decision Processes* 101, 2 (2006), 127–151.
[4] G. Booch et al. 2021. Thinking fast and slow in AI. In *Proc. AAAI*, Vol. 35, 15042–15046.
[5] L. Chen, M. Zaharia, and J. Zou. 2023. FrugalGPT: How to use large language models while reducing cost and improving performance. arXiv:2305.05176.
[6] X. Chen et al. 2024. Do NOT think that much for 2+3=? On the overthinking of o1-like LLMs. arXiv:2412.21187.
[7] C. Clark et al. 2019. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proc. NAACL*, 2924–2936.
[8] K. Cobbe et al. 2021. Training verifiers to solve math word problems. arXiv:2110.14168.
[9] W. De Neys. 2012. Bias and conflict: A case for logical intuitions. *Perspectives on Psychological Science* 7, 1 (2012), 28–38.
[10] J. St. B. T. Evans and K. E. Stanovich. 2013. Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science* 8, 3 (2013), 223–241.
[11] S. Frederick. 2005. Cognitive reflection and decision making. *Journal of Economic Perspectives* 19, 4 (2005), 25–42.
[12] A. P. Gema et al. 2025. Inverse scaling in test-time compute. *Transactions on Machine Learning Research* (2025). arXiv:2507.14417.
[13] Y. Geifman and R. El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in NeurIPS*, Vol. 30.
[14] B. Grofman, G. Owen, and S. L. Feld. 1983. Thirteen theorems in search of the truth. *Theory and Decision* 15, 3 (1983), 261–278.
[15] M. S. Hagger et al. 2016. A multilab preregistered replication of the ego-depletion effect. *Perspectives on Psychological Science* 11, 4 (2016), 546–573.
[16] TypeSafe / Jev AI. 2026. What is Jev? The TypeSafe System One model explained. https://jev-ai.pro/what-is-jev. Accessed Sept. 2026.
[17] D. Kahneman. 1973. *Attention and Effort*. Prentice-Hall, Englewood Cliffs, NJ.
[18] D. Kahneman. 2011. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York.
[19] D. Kahneman and S. Frederick. 2002. Representativeness revisited: Attribute substitution in intuitive judgment. In *Heuristics and Biases*, T. Gilovich, D. Griffin, and D. Kahneman (Eds.). Cambridge University Press, 49–81.
[20] D. Kahneman and G. Klein. 2009. Conditions for intuitive expertise: A failure to disagree. *American Psychologist* 64, 6 (2009), 515–526.
[21] D. Kahneman, O. Sibony, and C. R. Sunstein. 2021. *Noise: A Flaw in Human Judgment*. Little, Brown Spark, New York.
[22] D. Kahneman and A. Tversky. 1979. Prospect theory: An analysis of decision under risk. *Econometrica* 47, 2 (1979), 263–291.
[23] T. Lee. 2026. Thinking fast and slow with tree ensembles. Manuscript.
[24] F. Lieder and T. L. Griffiths. 2020. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences* 43 (2020), e1.
[25] C. G. Lord, M. R. Lepper, and E. Preston. 1984. Considering the opposite: A corrective strategy for social judgment. *Journal of Personality and Social Psychology* 47, 6 (1984), 1231–1243.
[26] P. Milgrom and C. Shannon. 1994. Monotone comparative statics. *Econometrica* 62, 1 (1994), 157–180.
[27] OpenRouter. 2026. What is Jev? TypeSafe's decision model explained for developers. Blog post.
[28] G. Pennycook, J. A. Fugelsang, and D. J. Koehler. 2015. What makes us think? A three-stage dual-process model of analytic engagement. *Cognitive Psychology* 80 (2015), 34–72.
[29] A. Yang et al. 2025. Qwen3 technical report. arXiv:2505.09388.
[30] Qwen Team. 2025. Qwen3-8B model card and quickstart. https://huggingface.co/Qwen/Qwen3-8B. Accessed Sept. 2026.
[31] M. Sadinle, J. Lei, and L. Wasserman. 2019. Least ambiguous set-valued classifiers with bounded error levels. *J. Amer. Statist. Assoc.* 114, 525 (2019), 223–234.
[32] A. Shenhav, M. M. Botvinick, and J. D. Cohen. 2013. The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron* 79, 2 (2013), 217–240.
[33] S. A. Sloman. 1996. The empirical case for two systems of reasoning. *Psychological Bulletin* 119, 1 (1996), 3–22.
[34] C. Snell, J. Lee, K. Xu, and A. Kumar. 2024. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. arXiv:2408.03314.
[35] V. A. Thompson, J. A. Prowse Turner, and G. Pennycook. 2011. Intuition, reason, and metacognition. *Cognitive Psychology* 63, 3 (2011), 107–140.
[36] D. M. Topkis. 1978. Minimizing a submodular function on a lattice. *Operations Research* 26, 2 (1978), 305–321.
[37] A. Tversky and D. Kahneman. 1974. Judgment under uncertainty: Heuristics and biases. *Science* 185, 4157 (1974), 1124–1131.
[38] E. Vul and H. Pashler. 2008. Measuring the crowd within: Probabilistic representations within individuals. *Psychological Science* 19, 7 (2008), 645–647.
[39] A. Wald. 1945. Sequential tests of statistical hypotheses. *Annals of Mathematical Statistics* 16, 2 (1945), 117–186.
[40] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *Proc. ICLR*.
[41] M. Xiong et al. 2024. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *Proc. ICLR*.